

Full Stack Development with MERN

Project Documentation

ServiceNow Flow Designer Implementation & Workflow Specification

Date: 1 October 2026	Team ID:
Project Title: Auto ticket classification using flow designer	Maximum Marks: 4 Marks

1. Introduction

This project introduces an automated solution for classifying and routing incoming IT support tickets within the ServiceNow platform using Flow Designer. By replacing manual triage with keyword evaluation and conditional decision rules, the system streamlines issue categorization and immediate group routing.

Team Members and Roles:

Team Member	Role / Responsibility
SUWETHA M	Team Lead — Architecture, Requirement Analysis & Coordination
PRIYADHARSHINI S	Team Member — Backend Data Configuration & Schema Setup
SWATHI P	Team Member — Flow Designer & Decision Logic Automation
JEBA ASHIKA X	Team Member — Validation, End-to-End Testing & Deployment

2. Project Overview

Purpose:

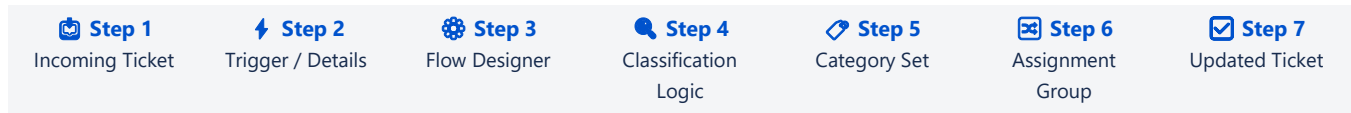
The purpose of the project is to automate the classification and routing of incoming support tickets, reducing manual effort and helping tickets reach the appropriate category and assignment group more efficiently.

Key Features:

- Automatic ticket classification based on incoming text details.
- Condition/decision-based classification using Flow Designer rules.
- Automatic assignment group routing to minimize delays.
- Flow Designer-based workflow automation operating without custom code.
- Reduced manual intervention and lower triage overhead.
- Handling of unmatched or exceptional tickets via fallback routing.
- Faster and more consistent ticket processing across helpdesk teams.

3. Architecture

The application architecture leverages ServiceNow's native event-driven engine. Upon ticket creation, the Flow Designer workflow captures record attributes, applies decision branch logic, sets appropriate Category and Subcategory values, and updates the Assignment Group.



4. Setup Instructions

Prerequisites:

- ServiceNow Personal Developer Instance (PDI) or Enterprise Instance.
- Access to Flow Designer application with admin or flow_operator roles.
- Required application/table permissions on the target Incident table.
- Appropriate user roles configured for assignment group members.

Configuration Steps:

1. Log in to the ServiceNow Instance as System Administrator.
2. Navigate to Process Automation > Flow Designer and create a new Flow.
3. Configure Trigger: Record Created on the target table where Category is empty.
4. Add Decision Logic Actions: Evaluate Short Description keywords (e.g., 'network', 'password', 'hardware').
5. Add Update Record Actions: Assign Category, Subcategory, and Assignment Group based on branch evaluation.
6. Add Exception Handling: Route unmatched records to Helpdesk Service Desk for manual triage.
7. Save and Activate the Flow Designer workflow.

5. Project Structure / Configuration Structure

In place of traditional code file directories, the ServiceNow Flow Designer application is structured into modular components within the instance database:

- **Flow Trigger:** Defines the record creation event and initial condition filtering.
- **Actions & Data Pills:** Extracts incoming ticket fields and passes context downstream.
- **Conditions & Decision Logic:** Evaluates string patterns and dictionary mapping rules.
- **Classification Rules:** Sets Category and Subcategory field values dynamically.
- **Assignment Group Rules:** Resolves target support team references based on category.
- **Exception / Fallback Handling:** Catches unclassified tickets and flags them for manual review.

6. Running the Application

Execution in ServiceNow occurs automatically when a support ticket event is triggered. Testing and execution follow this standard lifecycle:

1. Create / Receive Ticket: A user submits a new issue ticket via Portal or Incident Form.
2. Trigger Flow: ServiceNow detects record creation and invokes the active Flow Designer flow.
3. Classify Ticket: Decision logic scans text and assigns appropriate Category and Subcategory.
4. Route Ticket: The workflow maps the classification to the matching Assignment Group.
5. Update Assignment Group: Ticket record fields update instantly, and assigned technicians are notified.

7. Flow / Integration Documentation

The table below documents the core flow components and data actions utilized:

Component	Purpose
Trigger	Starts the flow when a ticket is created or updated.
Classification Logic	Determines the ticket category and subcategory.
Routing Logic	Selects the appropriate assignment group.
Update Action	Updates ticket fields directly on the target record.
Fallback Action	Handles unmatched tickets by routing to Tier 1 support.

8. Authentication & Access Control

Authentication and access controls are governed by ServiceNow Role-Based Access Control (RBAC). Workflow execution operates under the System User context or assigned service accounts, ensuring that ticket field updates strictly respect instance security policies and table permissions.

9. User Interface

Placeholder frames for ServiceNow platform interface screens:

- [Visual Placeholder: ServiceNow Support Ticket / Incident Input Form]
- [Visual Placeholder: Flow Designer Trigger and Decision Tree Configuration]
- [Visual Placeholder: Automatic Category & Subcategory Classification Outcome]
- [Visual Placeholder: Automated Assignment Group Routing & Notification Result]

10. Testing Matrix

Expected test results across defined ticket scenarios:

Test Case	Input / Scenario	Expected Result
Valid Classification	Ticket short description matches rule keywords	Correct category and subcategory assigned
Group Assignment	Ticket matches assignment group condition	Correct support team assigned
Unmatched Ticket	No classification rules match incoming description	Routed to Fallback / Manual Triage Desk
Incomplete Details	Required ticket fields or descriptions missing	Exception flag applied for review

11. Screenshots or Demo

Documented figure captions for instance verification:

- **Figure 1 – ServiceNow Ticket Creation Interface**
- **Figure 2 – Flow Designer Workflow & Trigger Configuration**
- **Figure 3 – Ticket Classification Rule Processing**
- **Figure 4 – Automatic Assignment Group Routing Verification**

Demo Link: [\[Insert ServiceNow Instance Demo / Video Link Here \]](#)

12. Known Issues & Limitations

- Tickets with highly ambiguous or non-standard descriptions may require manual review.
- Classification accuracy relies directly on the exhaustiveness of configured keyword rules.
- Unmatched ticket types trigger fallback routing and require human intervention.

13. Future Enhancements

- AI/ML-based ticket classification integration.
- Natural-language understanding (NLU) for complex ticket descriptions.
- Confidence scoring to trigger automated assignment thresholds.
- Machine learning models trained on historical ticket resolution data.
- Performance analytics dashboard to monitor classification accuracy.